

GeneyMAP: Exploring the Potential of GenAI to Facilitate Mapping User Journeys for UX Design

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Abstract

Generative AI (GenAI) has been widely applied in UX design, yet its potential in the Journey Map (JM) creation process remains under-explored. We conducted a formative study (N = 24) to identify designers' needs for GenAI in JM creation, resulting in six design goals (e.g., Acting as Different Stakeholders) implemented in our tool, GeneyMAP. GeneyMAP streamlines the JM creation process, allowing designers to map interview data efficiently with flexibility, uncovering design opportunities through visual inspiration. A subsequent user study (N = 20) demonstrated that GeneyMAP, compared with the common tool, accelerated JM creation and fostered creativity mainly by providing diverse inspirations and facilitating progressive discussions. Our findings proved GeneyMAP's utility and effectiveness while challenges in maintaining control and trust in GenAI outputs were noted. Our research highlights the promising role of GenAI in refining JM creation practices and suggests implications for incorporating GenAI in JM and design workflows.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**.

Keywords

Generative AI, LLMs, Design Tool, UX Design, User Research, Journey Map

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1 Introduction

Generative Artificial Intelligence (GenAI) applications, such as ChatGPT and Stable Diffusion, have made significant advancements in creating both textual and visual content. With the strength to process vast amounts of data [16], generate insights [52, 56] and produce visualizations [5, 62], GenAI has shown potential in design tasks like idea generation [8, 50], prototyping [11], and generating evaluation reports [71]. User Experience (UX) design, a key area of Human-Computer Interaction (HCI) research that focuses on enhancing the user's overall interaction with products or systems [46], stands to benefit from these advancements. Currently, UX design scholars are actively exploring how to effectively harness GenAI in the UX design process [61], aiming to boost productivity and creativity [68], reduce barriers to design [31], and facilitate creative collaboration [37]. For example, integrating GenAI into design tools like persona [57] and storyboard [39] could significantly reduce the time and resources required to develop and update UX design solutions.

Journey Map (JM) is a critical UX design tool that visually represents the actions users take when experiencing a product or service, helping designers create intuitive and engaging experiences that align with user expectations [60]. Existing commonly used JM tools (hereafter referred to as “common tools”) face notable challenges in managing large datasets required for comprehensive analysis and rely heavily on human interpretation of qualitative data (e.g., interview scripts). The rigid creation process provided by common tools may obscure valuable insights, potentially limiting the creativity and quality of JM outputs. Meanwhile, GenAI excels in synthesizing information [54, 57, 63], identifying opportunities [22], and creative ideating [1, 25], offering the potential to significantly improve key aspects of JM creation. However, a considerable gap remains in our

understanding of how to systematically leverage GenAI to *streamline* the JM creation process – specifically, to accelerate the creation process while preserving the integrity of JM workflow (i.e., distilling journey phases, mapping user data, identifying pain points and visualizing potential design opportunities) [63]. To bridge this gap, we explored the potential of GenAI in facilitating the creation of JMs and aimed to address three research questions (RQs). **RQ1:** *What expectations do designers have for GenAI in facilitating the creation of JMs?* **RQ2:** *What is the impact of GenAI on the process and outputs of creating JMs, particularly in comparison to the common tool?* **RQ3:** *How do designers perceive the incorporation of GenAI in creating JMs, particularly in comparison to the common tool?*

To answer the research questions, we have followed the Research through Design paradigm [76] by developing and evaluating a GenAI-empowered system for creating journey maps named **GeneyMAP** with designers. Firstly, we conducted a formative study involving eight design practitioners, eight educators, and eight design students. Through semi-structured interviews, we identified their expectations for GenAI in the JM creation process, providing insights into how GenAI could facilitate JM development (**RQ1**). Secondly, six experienced design experts participated in the co-design of GeneyMAP, which involved creating a low-fidelity prototype and conducting heuristic evaluations and iterations. GeneyMAP can distill the JM templates and map user data automatically with tailored GenAI agents, enabling the exploration of visualized inspirations. Lastly, we validated GeneyMAP through a comparative evaluation against a chosen common tool. Twenty designer participants used both the common tool and GeneyMAP to complete two distinct JM creation tasks.

Throughout the evaluation, we collected the following post-trial data to answer **RQ2** and **RQ3**: 1) the time recorded to complete each JM output with GeneyMAP/the common tool; 2) expert assessment (i.e., CAT questionnaire) on the JM outputs generated by GeneyMAP/the common tool; 3) Self-assessment of the effectiveness of two tools throughout the entire JM process; 4) CSI questionnaires, focusing on the perceived creative support from GeneyMAP/the common tool; 5) customized questionnaires, based on previous research, aiming to assess the agency of two tools; 6) NASA-TLX questionnaires for participants to share their ratings on perceived usability of using two different tools; 7) semi-structured interviews that were focused on participants' perceptions regarding the roles of GeneyMAP/the common tool and their reasoning behind the ratings. Through quantitative and qualitative analysis of data collected via time recordings, questionnaires, and interviews, we found that GenAI improved JM outputs in terms of *novelty*, *thoughtfulness*, *interestingness*, and *surprise*. Additionally, it accelerated the JM creation process, with strengths such as “*comprehensive JM phases distillation*” and “*efficient user data mapping*” (**RQ2**). Simultaneously, participants' perceptions revealed how they viewed GenAI's role in JM creation, highlighting the perception of creativity support (e.g., “*providing insightful stimuli*” and “*lacking originality*”), perception of agency (e.g., “*empowering designer authority*” and “*challenges in controlling details*”) and perception of usability (e.g., “*enabling flexible modification*”, and “*required manual verification*”) (**RQ3**).

Our research on incorporating GenAI into the JM creation process contributes to the field of design, especially UX design, in

three primary ways. 1) To our knowledge, this is the first study that has explored the integration of GenAI throughout the entire JM creation process, from analyzing user data to mapping information, identifying pain points, and ideating design opportunities. 2) We provide empirical insights about the advantages, limitations, and strategies of incorporating GenAI in the JM creation process through a comparative analysis between GeneyMAP and the common tool, from the perspective of designers. 3) We highlight the challenges and opportunities derived from the empirical findings for empowering designers to collaborate with GenAI efficiently, suggesting the design implications for the development of future GenAI-empowered JM and design tools.

2 RELATED WORKS

2.1 Journey Maps in UX Design

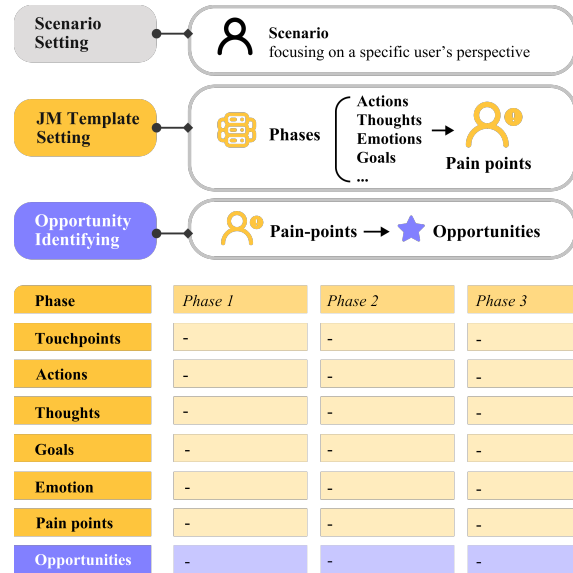


Figure 1: The typical template and its building workflow of a JM.

JM is an essential visualization design tool in UX design [35] that captures users' interactions with a system over time, which is not only taught widely in design education but also commonly applied in corporations to understand user behavior. By mapping the dynamic flow of user experiences, JMs provide rich insights into interactions and the broader context in which these experiences take place [24], offering a comprehensive view across multiple UX phases [15, 59].

The JM creation process begins with data collection, utilizing methods such as interviews, surveys, direct observations, and ethnographic studies to capture various facets of the UX [60]. **User Interviews** are commonly used to collect direct feedback, which is then analyzed to identify patterns in user behavior and derive valuable insights into the user journey [28]. Once data is collected, designers proceed to map the user journey, generally following a three-step process [29] (see Figure 1). The first step involves defining the

Scenario, focusing on the specific user's perspective and the contextual framework of their **goals** and expected outcomes [15]. In the second step, by detailing **Phases** along with **Actions**, **Thoughts**, **Emotions**, and **Goals**, designers can capture the sequential nature of user interactions, then map out behavioral steps and emotional responses throughout the journey [24, 59]. By mapping out the user experience, designers can recognize key user interactions with the system, facilitating a deeper understanding of their needs and behaviors [26]. The final step is to identify **Pain Points** and **Opportunities** from the mapping process to translate insights into actionable strategies for improving the UX [44, 64]. By illustrating the pain points and opportunities revealed within the JM, designers can foster a shared vision and promote meaningful dialogue among team members and stakeholders [35].

Despite their widespread use and benefits, existing JMs face significant challenges in managing large datasets necessary for comprehensive analysis. Another critical challenge is the objective interpretation of user data, particularly interview data, where biases in interpretation can hinder the identification of pain points. Our research aims to explore the potential of GenAI in facilitating the creation of JMs, addressing the challenges in terms of data interpretation, and inspiration generation.

2.2 Generative AI Applications for Supporting UX Design

The HCI community has shown significant interest in the intersection of GenAI and UX [17, 45], with applications spanning various stages of the UX design process for improving productivity [27, 33], and enhancing creativity [5]. Text-to-text (e.g., ChatGPT) and text-to-image (e.g., Dall-E) applications are the most discussed types of GenAI in UX design, which are applied to tasks such as persona creation [43, 54] and prototyping [38].

Text-to-text GenAI applications, powered by Large Language Models (LLMs), are particularly effective in the early stages of UX design, such as desktop research [73] and ideation [14, 67, 74]. Leveraging vast textual training data and human feedback, these models excel at tasks such as text summarisation [21], automating repetitive tasks with simple logic [23, 32], and analyzing qualitative user data [16, 50]. Moreover, LLMs can help extract key insights from user stories [52] or code qualitative interviews [42]. **Text-to-image** GenAI applications are particularly useful in UX design stages requiring conceptual visualization and prototyping, such as sketching [30, 47], wireframing [3, 11], and layout optimization [7, 65]. Text-to-image applications enable designers to generate conceptual images [55] and interactive UI mock-ups [69] using natural language inputs, significantly speeding up the rapid iteration of visual concepts with minimal manual effort [63].

Despite the strengths of GenAI applications in enhancing UX design, several challenges remain and require further exploration. For example, "hallucinations" (i.e. factually incorrect outputs) generated by GenAI can erode trust in the content it produces [6, 75]. Additionally, the illogical collation of input data [19, 68] caused by a lack of capabilities for multi-step reasoning [70] may deepen the mistrust and reduce design efficiency. The quality of GenAI outputs also depends heavily on the input prompts [41], necessitating

thoughtful prompt engineering. Manual prompts are often suboptimal, while tool-assisted prompts tend to be less human-readable [58].

The potential of GenAI to support the JM creation process is gaining recognition, as GenAI can process user interview scripts and generate insightful ideas. Existing solutions such as QoQo.ai [51], which rely on LLM to synthesize user journeys without real-world user input, raise concerns about the authenticity of generated content. Given that the potential of GenAI in assisting JM creation has been largely underexplored, our research aims to address the gap through the development of a GenAI-empowered JM tool that supports both automatic generation based on interview scripts and manual input, allowing designers to customize outputs while benefiting from AI-driven insights.

3 FORMATIVE STUDY

3.1 Method

To derive the design goals for developing a GenAI-empowered journey mapping tool, we conducted semi-structured interviews complemented by thematic analysis, identifying how designers envisioned the integration of GenAI into their JM creation. Participants from both academic institutions and industry settings were recruited to gather multi-stakeholder perspectives. The interviews were conducted online between January and April of 2024.

3.1.1 Participants. Twenty-four design-trained participants (15 males, 9 females) were recruited, with an average age of 27.79 ($SD = 4.34$). The participants comprised eight practitioners, eight students, and eight educators, all practicing in the field of UX design, although they had varying educational design backgrounds. Recruitment began through the authors' professional networks and social media and later expanded through snowball sampling techniques. The participants were drawn from various industries, including electronics, robotics, and automotive sectors, along with academic institutions known for their expertise in interaction design. The demographic information of participants is presented in Appendix - A. All participants had practical experience with GenAI tools, such as text-to-image platforms and LLM services, which they had integrated into their design workflows. Each interview, conducted in the participants' preferred language (i.e., Mandarin Chinese) to facilitate discussion, ranged from 40 to 70 minutes with an incentive valued at CNY 100.

3.1.2 Interview Procedure and Data Analysis. Before each session, the interviewer introduced the study topics and explained data protection measures to comply with ethical standards. With participant consent, all interviews were audio-recorded. During the sessions, participants discussed their experiences using GenAI in their design workflows, with a focus on JM creation. The interview included reflections on the adaptations necessary to integrate GenAI, efficiency gains, any challenges encountered, and the expected functions of GenAI in their future JM creation process. Interviewers actively posed in-depth follow-up questions based on initial responses to ensure a comprehensive exploration of each topic.

The interview data were transcribed and analyzed using an inductive thematic approach. The analysis focused on 'Design Goals' corresponding to the **RQ1**, outlining participants' expectations for

using GenAI in JM creation and their perceptions of its impact on their workflows. Two researchers independently coded the data and then discussed and refined the codes to ensure consistency and comprehensiveness. A third researcher validated and confirmed the final themes. For clarity in reporting findings, participants were designated with labels: I1-I8 for industrial practitioners, E1-E8 for educators, and S1-S8 for students (see Figure 2).

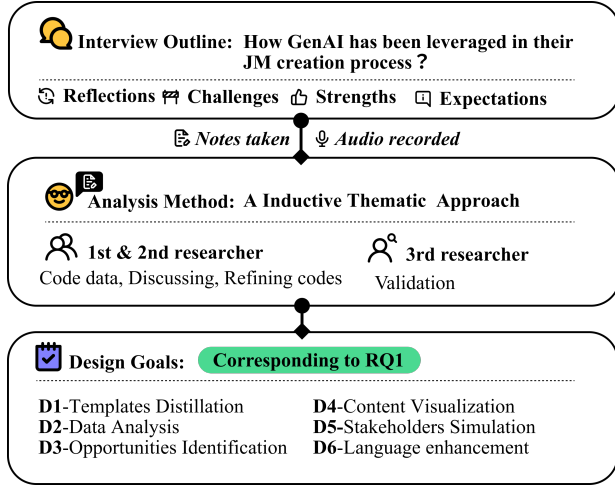


Figure 2: The interview procedure and data analysis methods of formative study.

3.2 Design Goals

This section synthesizes insights to address **RQ1**. The following Design goals (**D1-D6**) were derived from interview scripts regarding how GenAI could be effectively utilized throughout the JM creation process.

D1 - Structured JM template distillation. GenAI can assist designers by quickly generating structured templates, particularly in identifying user journey phases. S2 suggested that GenAI would be particularly valuable to “merge relevant themes into a focused theme”. GenAI’s ability to customize structures allows for seamless integration into common JM workflows, where templates serve as foundational frameworks that typically require manual creation by designers. **D1** highlights the transformative potential of GenAI in streamlining JM creation processes.

D2 - User data analysis and mapping. GenAI can enable designers to rapidly grasp vast amounts of textual information by facilitating the analysis and summarisation of user data, which saves time and enhances accuracy. For example, S4 shared that GenAI could “map user activities to the JM template rapidly, and we could immediately optimize the content based on what has been organized”. Additionally, GenAI is expected to handle lengthy user interviews by providing “concise summaries that retain essential insights” (I2), for “the fewer words on a JM, the easier it is to focus on key design points” (I5).

D3 - Design opportunities identification. GenAI is pivotal in identifying and exploring design opportunities. By quickly gathering textual information and generating various ideas based on

the revealed pain points, GenAI enables designers to identify new design opportunities efficiently (I1). Additionally, **D3** is further enhanced by GenAI’s ability to offer diverse perspectives that support the iteration and refinement of design concepts, broadening exploration and revealing untapped potential (e.g., E4, E2).

D4 - Insightful visual content generation. GenAI is expected to facilitate the generation of visualized materials that cater to communication and inspiration. In terms of expression, images generated by GenAI aid designers in presenting initial concepts without requiring extensive visualization skills (e.g., I2, I8), improving the communication of design concepts where visual representations are paramount. For inspiration, GenAI can provide inspiring visuals, aiding in digging into potential user interactions and solutions (e.g., S8, E2). **D4** promotes an efficient and comprehensive approach to design exploration and realization.

D5 - Acting as different stakeholders. GenAI offers the unique ability to simulate various stakeholders, acting as users, design experts, etc. This capability allows GenAI to be “interviewed” as users, providing insights into user needs, or to assume the role of other stakeholders, offering immediate suggestions and evaluations (S7, S4). Additionally, I6 noted that ChatGPT can “identify design issues as an expert-level designer, suggesting potential strategies”. **D5** enables designers to effectively explore and address potential design challenges from multiple angles.

D6 - Language enhancement. GenAI plays a crucial role in enhancing expression within the JM process (e.g., describe a complex user action with few words), helping designers concisely describe intricate user actions (S8). Additionally, GenAI can reorganize colloquial explanations into logical descriptions and simplify complex terminologies with intuitive examples (I6). **D6** restructures the information in a comprehensible and accessible way, enhancing the effectiveness of communication within and beyond the design team (I7).

4 CO-DESIGNING GeneyMAP

4.1 The co-design process

Table 1: The demographic information of the experts who participated in the GeneyMAP co-design session.

	Gender	Age	JM Exp.	GenAI Exp.	Frequency of Using GenAI
Practitioner 1	F	25	7 years	20 months	Frequently
Practitioner 2	F	30	10 years	16 months	Frequently
Researcher 1	F	23	6 years	20 months	Frequently
Researcher 2	F	24	6 years	19 months	Always
Educator 1	M	31	12 years	30 months	Always
Educator 2	F	29	12 years	24 months	Frequently

1. Always: participant utilizes GenAI in every design project. 2. Frequently: participant utilizes GenAI in the majority of design projects.

The co-design process of GeneyMAP incorporated direct insights from six design experts familiar with GenAI. The experts included two practitioners, two researchers, and two educators, all with at least six years of experience in integrating JMs into their design processes (the demographic information could be found in Table 1). The Co-design process began with an initial low-fidelity prototype, which was shaped by the design goals from **Section 3.2**. We conducted weekly meetings to evaluate the development progress of GeneyMAP, collecting hands-on feedback aimed at enhancing usability and professional applicability. We systematically incorporated the feedback into the refinement of GeneyMAP, aligning it closely with designer expectations. After the iterative revisions, a heuristic evaluation was conducted by the six design experts to ensure that the final version of GeneyMAP met the design goals **D1-D6** (see Figure 3).

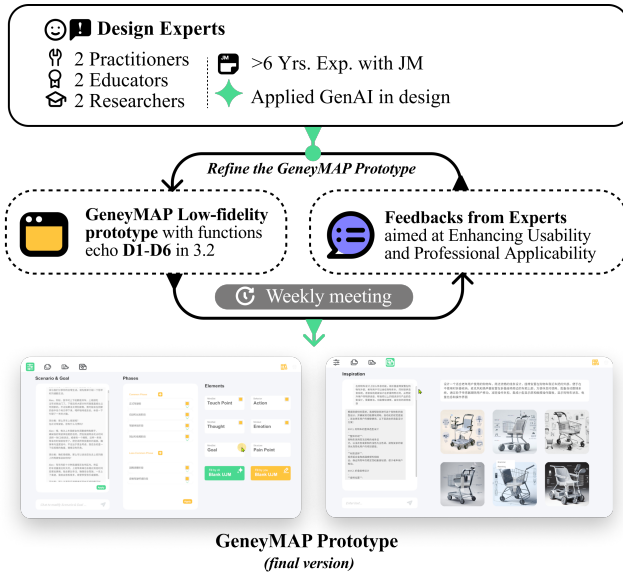


Figure 3: The co-design process of GeneyMAP.

4.2 Description of GeneyMAP

Developed using the Unity engine, GeneyMAP integrates OpenAI’s GPT-4o [49] and DALL-E 3 [48] APIs to enable textual data analysis and visualization for JM creation. GeneyMAP utilizes a unified JSON format to efficiently manage data exchange, ensuring the accurate generation of JM elements. A robust UI management system supports dynamic updates with features like undo, collapsible sections, and GenAI-driven text optimization. Additionally, the Runtime File Browser [66] plugin facilitates seamless file operations within the Unity environment.

GeneyMAP provides a progressive four-step JM creation process, designed to fulfill the expected design goals (D1-D6). Figure 4.b presents an overview of the workflow in pseudocode, outlining the sequence of operations within each step. The figure also highlights the automated GenAI functions with comprehensive explanations for each operation. GeneyMAP’s prompt engineering, which is

rooted in a chain-of-thought (CoT) approach, systematically transforms explicit rules into operational prompts to implement the system workflow. The CoT begins by defining the background of the design task, JM terminologies (i.e., scenario, phase, touchpoint, action, emotion, pain points, and opportunities), JM creation process, and GenAI’s responsibilities. These foundational rules serve as consistent principles that govern the JM creation workflow across all four steps, ensuring coherence with the intended design goals. To operationalize these rules, illustrative examples are embedded into the prompts to clarify the relationship between input and output. Finally, the prompts explicitly emphasize the required output structure, guaranteeing compatibility with the subsequent steps of the JM creation. A sample prompt of how GeneyMAP generates opportunities from the selected pain points is illustrated in Figure 4.a.

While GenAI automates most steps, GeneyMAP integrates prompt-based interaction as a flexible feature to support diverse needs, enabling designers to refine outputs and maintain quality control. This design choice not only accommodates designers’ familiarity with existing GenAI tools but also balances efficiency with control, catering to varying levels of designer involvement. The detailed use scenario of the four JM creation steps is outlined below (see Figure 5).

Step 1. Distilling the Structured JM Template. This step aims to create a structured foundation for JM creation, ensuring alignment with user insights. Building on the JM structure mentioned in Section 2.1, GeneyMAP addresses **D1-Templates Distillation** by allowing designers to upload all interview scripts, which serve as the data source for identifying a scenario and phases. GeneyMAP employs LLMs to analyze the scripts, summarizing the scenario by distilling a sequence of events into concise descriptions. Designers can refine the auto-generated scenario through interactive prompts with GenAI or manually edit the scenario. The scenario is embedded in the system prompt for all subsequent GenAI functions. Guided by the refined scenario, statistical analysis powered by LLMs identifies frequently occurring phases as common JM phases and less frequent ones as less common phases. These phases are reviewed by designers to form the horizontal axis of the JM template. Finally, designers select elements including touchpoints, actions, emotions, thoughts, goals, and pain points to construct the vertical axis of the JM template, ensuring that all relevant aspects of the UX are captured (see Figure 5.a). Once the template is set up, designers can either (1) click the “Fill by AI” button to have GenAI automatically map information, or (2) create a blank template for manual input.

Step 2. Generating Individual JMs. This step aims to transform raw interview data into structured JMs that represent individual user experiences while aligning with the predefined scenario. By leveraging LLMs for efficient script processing, GeneyMAP automates the mapping of user-specific insights to the established templates (echoes **D2-Data Analysis**). Each interview script is mapped to one real-user JM, ensuring a comprehensive representation of the real user’s journey. Additionally, GeneyMAP provides a “synthetic user function” aligned with **D5-Stakeholders Simulation**, enabling GenAI to simulate users based on the set scenario, which generates alternative JMs that provide additional insights (see Figure 5.b). The synthetic user is an optional function designed to offer additional inspirational stimuli rather than being a mandatory step for

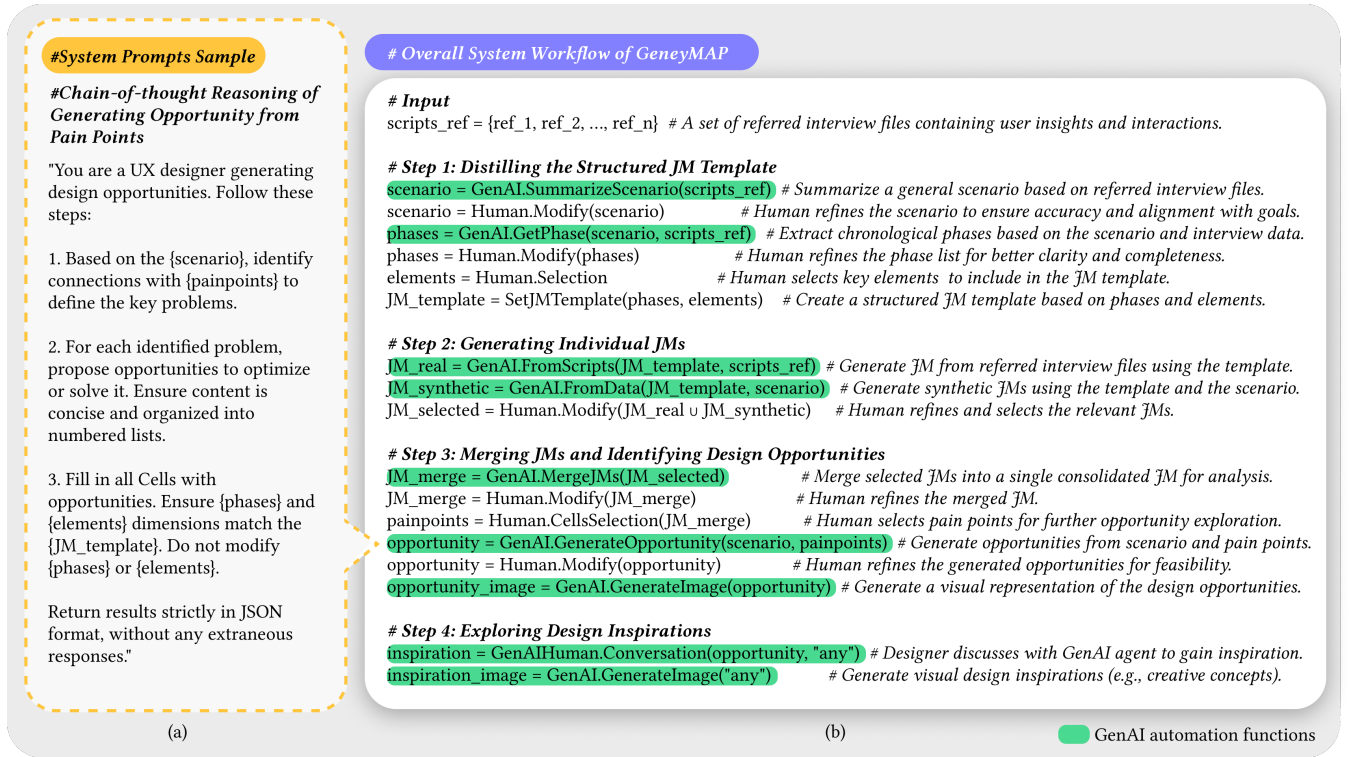


Figure 4: The system workflow and system prompts sample of GeneyMAP.

completing a JM. Designers can refine the JMs by adding or removing rows, columns, and text content, leveraging GenAI's language processing capabilities to achieve **D6-Language Enhancement**.

Step 3. Merging JMs and Identifying Design Opportunities. This step focuses on merging individual JMs to create an overview and uncover actionable design opportunities. By clicking the "Merge" button, GeneyMAP is prompted to identify overlapping insights and condense similar content into concise summaries. This merging process strictly includes authentic insights from real-user JMs, avoiding fabricated data. Each cell in the merged JM can be selected to serve as hints for generating two opportunities automatically with corresponding images, thereby echoing **D3-Opportunities Identification** and **D4-Content Visualization**. Designers can generate or write more opportunities when needed (see Figure 5.c). Following this process, GeneyMAP is prompted to identify opportunities by isolating pain points within their respective phase, analyzing their root cause, and linking them to the scenario. Guided by this analysis, GeneyMAP is supposed to systematically generate actionable, contextually relevant design opportunities, while maintaining alignment with the JM structure (see Figure 4.a).

Step 4. Exploring Design Inspirations. This step focuses on discussing and ideating design inspirations through an interactive prompt-based GenAI agent. The agent utilizes pre-configured prompts to ensure contextual awareness of design tasks, eliminating the need to repeatedly explain the context. The agent facilitates articulating complex design concepts and reorganizing them into clear descriptions, enhancing communication within the design

team. By simulating expert feedback and providing iterative suggestions, GeneyMAP supports designers in refining their concepts and exploring innovative solutions (echoes **D5-Stakeholders Simulation**). Moreover, GeneyMAP can assist in crafting prompts for image generation, enhancing the quality of visual outputs to serve as creative design inspiration (echoes **D4-Content Visualization**, see Figure 5.d). Additionally, **D6-Language Enhancement** is integrated as a fundamental function throughout all four steps, assisting in articulating and refining textual content across the JM creation process.

5 USER STUDY

To investigate the impact of GenAI on the JM creation process and outputs, and assess the creativity, agency, and usability of GenAI in supporting the JM, we conducted a comparative user study involving both GeneyMAP and a common tool, which served as a baseline.

5.1 Participants

We recruited 20 participants (7 male, 13 female) through our professional networks, including acquaintances and referrals from the design community. To avoid any potential bias, no participants from the formative study were included in the user study. Participants were required to have at least five years of experience in the field of UX design, a relevant educational background, and prior experience with integrating GenAI into their design practice. The participants, aged from 22 to 29 ($M = 25.10$, $SD = 2.34$), were either UX-related

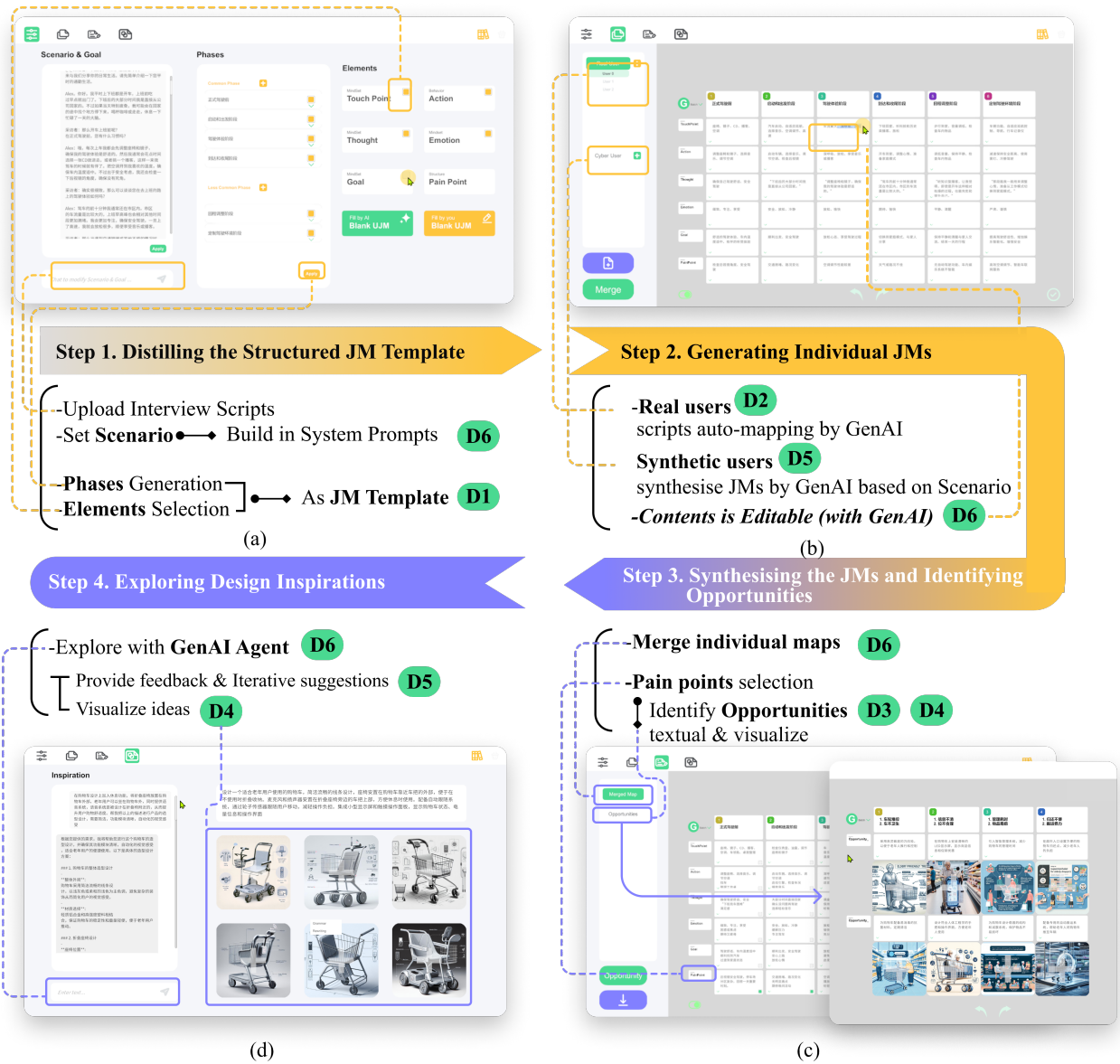


Figure 5: The detailed use scenario of the four JM creation steps with GeneyMAP.

designers or UX design graduates (see Table 2). The study was conducted in person, and each session was video-recorded, including screen recordings of the tool use. After completing the test, participants were compensated with a shopping coupon valued at 200–300 HKD, depending on the length of their participation.

5.2 Procedure

We conducted the user study using a within-subjects design over a 2.5-hour duration. Prior to the study, we had a 10-minute preliminary interview with each participant to gather background information. Participants were then instructed on how to use GeneyMAP, with a guidebook and a brief demonstration developed by the researchers. Following this, participants were tasked with creating

user journey maps using both the common tool and GeneyMAP. The common tool is a JM template without any GenAI functions, provided by the Figma platform. This template organizes JM phases horizontally, with elements such as touchpoints, actions, emotions, thoughts, goals, and pain points arranged vertically (see Figure 7 in Appendix - B), making it similar to GeneyMAP and the typical template presented in Figure 1. We selected this template for two key reasons: First, Figma is the most widely used platform among UX designers, and the adopted official template has been widely employed in numerous UX design projects [10]. Second, as outlined in RQ2&3, this study aims to explore the impact of GenAI on the JM creation, as well as designers' perceptions. Therefore, a blank JM template served as a clear baseline that allowed participants to

Table 2: The demographic information of the participants recruited for the user study.

No.	Gender	Age	Design Exp.	JM Exp.	GenAI Exp.	Frequently
P1	M	29	Interaction Design Researcher	7 years	24 months	Always
P2	F	28	Interactive System Design, UX/ UI Designer	9 years	12 months	Seldom
P3	F	27	Product Manager	5 years	6 months	Seldom
P4	M	23	Intelligent Interaction Design	5 years	19 months	Occasionally
P5	F	24	UX/ UI Designer, Product Designer	6 years	6 months	Seldom
P6	M	26	Interaction Design, UX Designer	8 years	24 months	Frequently
P7	F	24	Industrial Design, Intelligent Interaction Design	5 years	6 months	Seldom
P8	M	23	Intelligent Interaction Design	5 years	22 months	Occasionally
P9	F	23	Industrial Design, Intelligent Interaction Design	5 years	15 months	Occasionally
P10	F	24	Interaction Design, UX/ UI Designer	6 years	13 months	Occasionally
P11	F	25	Intelligent Interaction Design, Product Manager	7 years	10 months	Occasionally
P12	F	29	Intelligent Interaction Design, UX/ UI Designer	6 years	18 months	Frequently
P13	F	22	Industrial Design, Intelligent Interaction Design	5 years	15 months	Seldom
P14	M	23	Interactive Media Design, Interaction Design	5 years	24 months	Always
P15	M	24	UX/ UI Design, Business Design	6 years	19 months	Occasionally
P16	M	27	Product Manager	5 years	16 months	Occasionally
P17	F	26	Interaction Design, UX/ UI Designer	6 years	12 months	Occasionally
P18	F	23	Interactive System Design	5 years	6 months	Seldom
P19	F	29	Intelligent Design Researcher	9 years	19 months	Seldom
P20	F	23	Intelligent Interaction Design, UX/ UI Designer	5 years	12 months	Occasionally

1. Always: Participant utilizes GenAI in every design project. 2. Frequently: Participant utilizes GenAI in the majority of design projects. 3. Occasionally: Participant utilizes GenAI in about half of their design projects. 4. Seldom: Participant utilizes GenAI only in several design projects. GenAI in the majority of design projects.

intuitively compare the presence or absence of GenAI functions, explicitly demonstrating the difference brought by GenAI. Two sets of user interviews *Scripts* were provided: *Scripts I*, related to the driving experience, and *Scripts II*, related to the shopping experience. The *Scripts* presented in Mandarin Chinese were constructed in conversation form to accommodate our Chinese-speaking participants (see Appendix - C).

Participants were divided into four groups (A, B, C, D), and we used a Latin square design to randomize the order of tools and scripts used across groups to mitigate any bias due to sequence effects (see Figure 6). For example, Group A used the common tool with Script I followed by GeneyMAP with Script II, while Group B first used the common tool with Script II and then GeneyMAP with Script I. This design ensured that each tool-script combination was tested under similar conditions and that all participants interacted with both tools and scripts.

Each session using GeneyMAP or the common tool lasted a maximum of 40 minutes, with a 15-minute break between sessions. After completing each session, participants filled out a set of questionnaires to gather immediate feedback on their experiences. Following both sessions, a semi-structured interview was conducted to collect participant feedback and understand the

reasons behind any rating differences. Finally, we conducted an expert review to assess the generated JMs from two tools. The data collection process is detailed in Section 5.3.

5.3 Data Collection

After using each tool, we recorded the *time taken* to complete each JM output. Participants then completed 10-point Likert-scale questionnaires to assess GeneyMAP and the common tool, with ratings ranging from 1 (the lowest performance) to 10 (the highest performance). The participants assessed their JM creation process by leveraging two different tools, using the process evaluation criteria from prior research [24] (see Table 3). Next, the Creativity Support Index (CSI) [4] was administered to assess how each tool facilitated *creative* tasks, helping us understand the extent to which each tool supported the creative JM process. In addition, we customized additional questions inspired by the work of Wu et al. [70] and Choi et al. [5] to assess the perceived *agency* of GenAI across the two tools, including transparency, control, and trust, etc. Lastly, we used the NASA Task Load Index (NASA-TLX) [20], a comprehensive measure of the perceived effort and cognitive load, providing assessments of the *usability* of different tools. These questionnaires

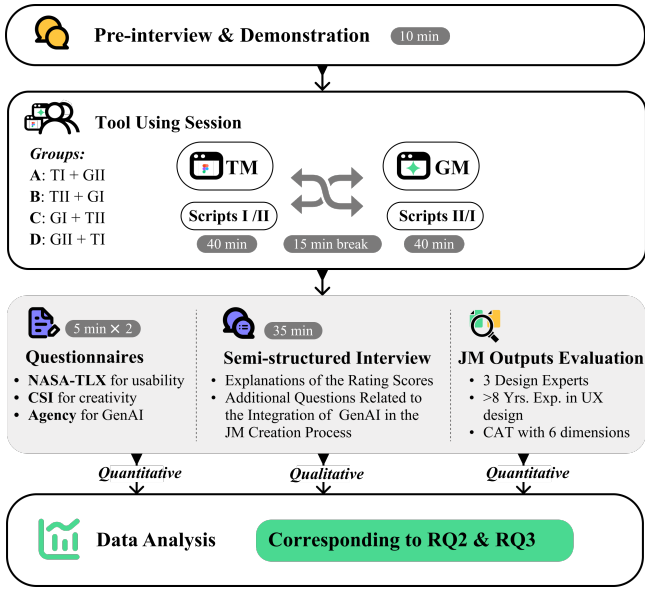


Figure 6: The procedure of user study.

collectively provided a robust and comprehensive framework for quantifying participants' experiences and perceptions.

After the two sessions, we conducted a *semi-structured interview* lasting 60-80 minutes with each participant to gather insights into the strengths and weaknesses of GeneyMAP and the common tool. Participants discussed what worked well, the challenges they encountered, and the reasons for any differences in their ratings of the two tools (e.g., why they considered GeneyMAP to be more or less effective in data mapping). Furthermore, expert evaluations of the JM outcomes were conducted by three experienced designers with at least eight years of UX design experience. These experts assessed the generated JMs (see Appendix - D) using a 10-point Likert scale questionnaire based on the **Consensual Assessment Technique (CAT)** [18], which evaluates six dimensions: novelty, appropriateness, thoughtfulness, interestingness, usefulness, and surprise.

5.4 Data Analysis

To answer **RQ2** concerning the impact of GenAI on the JM process and its outputs, we initially assessed interrater reliability among the questionnaire ratings of the three experts by conducting an intraclass correlation coefficient analysis [34], which achieved a score of .726 (*Moderate*). We then examined the distribution of data from time taken, CAT assessment, and self-assessment to determine the appropriate statistical method. A Shapiro-Wilk test confirmed that the data were normally distributed, so we conducted paired sample t-tests to compare time spent, expert CAT evaluations of the outputs, and self-assessment ratings between GeneyMAP and the common tool. At last, we conducted a thematic analysis of the transcribed interview data to identify the underlying reasons for the differences observed in the ratings. The analysis process led to the discovery of key themes, including *Comprehensive JM*

	Phases	It is easy to identify phases using the GeneyMAP/ common tool.
	Mapping	It is easy to map out each component (e.g., actions, emotions, thoughts) using the GeneyMAP/ common tool.
JM Process	Opportunities	It is easy to identify the pain points or the opportunities using the GeneyMAP/ common tool.
	Inspiration	It is easy to ideate the potential solution using the GeneyMAP/ common tool.
	Completeness	I am confident about the completeness of the user journey map created with the GeneyMAP/ common tool.
	Goal	The GeneyMAP/ common tool helped me think through what kinds of outputs I would want to complete, and how to complete the task.
	Transparency	The GeneyMAP/common creation system is transparent about how it arrives at its final result; I could track its progress.
Agency	Control	I felt I was controlling the GeneyMAP/ common tool to steer the system towards the task goal.
	Trust	I felt I trusted the outputs that were created through the use of the GeneyMAP/ common tool.
	Collaboration	I felt I was collaborating with the GeneyMAP/ common tool to come up with the outputs.

Table 3: The questionnaire customized to assess the JM creation process and Agency.

phases distillation", *Efficient user data mapping*", and *"Improving the Completeness"*, as detailed in Section 6.1.

To answer **RQ3** regarding designers' perceptions of GenAI in JM creation, we examined the data presenting the participants' perspectives on applying the GeneyMAP and common tool in the JM creation (i.e., interview data) and their quantitative assessment of the two tools (i.e., questionnaire data). A Shapiro-Wilk test confirmed that the data from the CSI, Agency, and NASA-TLX questionnaires were normally distributed, allowing us to use paired sample t-tests to compare the ratings for GeneyMAP and the common tool. The results revealed differences in perceived **Creativity Support** (Section 6.2), **Agency** (Section 6.3), and **Usability** (Section 6.4).

We then employed a mixed deductive-inductive method to analyze the interview data. Deductively, we used the evaluation dimensions (i.e., the perceived creativity support, agency, and usability) from the JM process, CSI, Agency, and NASA-TLX questionnaires as a framework to categorize participants' perceptions of using GeneyMAP compared to the baseline tool. Inductively, we explored both the positive and negative aspects of participants' experiences with GeneyMAP in the JM creation process, leading to the identification of themes such as *"facilitating progressive discussions"*, *"aligning with expectations"*, and *"over-reliance on GenAI"*. At the start of the analysis process, two researchers independently engaged with the interview data, identifying potential themes and creating initial codes. Regular meetings ensured consensus on the

Dimension	Themes	Participants
6.1 JM Process and Outputs	Comprehensive JM phases distillation	P6,P7,P11,P12
	Efficient user data mapping.	P1,P3,P4,P9,P10,P12,P13,P17
	Transforming pain points into opportunities.	P3,P11,P15
	Aiding in idea iteration.	P1,P3,P8,P13
	Improving the completeness of JMs.	P9,P10,P11,P17
	Inaccurate information mapping.	P4,P7,P13
	Irrelevant design opportunities.	P10,P14,P16
	Limited design engagement.	P7,P8,P11,P13
6.2 Perception of Creativity Support	Providing insightful stimuli.	P7,P9,P10,P11,P13
	Facilitating progressive discussions.	P3,P9,P11,P14
	Lacking originality.	P7,P10,P14
	Demanding effective prompts.	P1,P8,P12
6.3 Perception of Agency	Enhancing Objectivity.	P1,P7,P11,P13
	Empowering designer authority	P6,P7,P9,P11,P13
	Unexplainable reasoning.	P6,P9,P11,P12,P13
	GenAI-dominated output quality.	P6,P7,P10,P15
	Challenges in controlling details.	P10,P11
	Over-reliance on GenAI.	P9,P12,P13
6.4 Perception of Usability	Aligning with Expectations.	P1,P5,P6,P7,P9,P11,P12,P13,P15
	Enabling flexible modification.	P1,P9,P12,P16
	Improving communication clarity.	P6,P9,P14,P15
	Requiring manual verification.	P6,P17,P20,P21
	Refining unsatisfied output.	P8,P12

Table 4: The outline of the qualitative findings from the user study.

themes, with a third researcher resolving any coding discrepancies. To further validate our analysis, we randomly selected 10% of the interview data for analysis by a fourth researcher. The coding results from this fourth researcher were compared to those agreed upon by the initial three researchers, yielding a Cohen’s Kappa score of .746 (“Substantial” [36]). Ultimately, the qualitative analysis revealed the empirical findings that are summarized in Table 4.

6 FINDINGS

6.1 JM Process and Outputs

As shown in Table 5, the results of *time taken* showed that completing a JM using GeneyMAP ($M = 30.72$, $SD = 14.591$) took less time than with the baseline ($M = 36.63$, $SD = 4.477$, $p = .036$). Moreover, the two-sample pairwise t-test result of the CAT assessment showed that participants produced outputs with higher *Novelty* ($p = .016$), *Thoughtfulness* ($p = .020$), *Interestingness* ($p = .006$), and *Surprise* ($p = .007$) with the support of GeneyMAP than with the baseline.

Subsequently, we conducted paired t-tests to compare participants’ ratings for their self-assessment of the JM process. In terms of *phase* distillation ($p = .015$), user data *mapping* ($p < .001$), *opportunities* ($p = .048$) and *inspiration* generation ($p = .004$), and *completeness* ($p = .046$), participants rated GeneyMAP higher than the baseline. The rationales derived from participant interviews explained the rating differences are presented in the following sections, highlighting several strengths of GeneyMAP at different stages of the JM creation process:

- **Comprehensive JM phases distillation.** Participants appreciated GeneyMAP ($M = 8.10$, $SD = 2.360$) for its ability to “comprehensively extract JM phases based on my initial intuition” (P12), significantly outperforming the baseline ($M = 5.90$, $SD = 2.198$, $p = .015$). Even those who were skilled at the JM phase distillation (e.g., P6, P11) praised the tool’s thoroughness, and P7 noted that “these generated phases could be customized by cutting down to fit needs.”
- **Efficient user data mapping.** Compared to the baseline ($M = 4.55$, $SD = 2.502$), GeneyMAP ($M = 8.00$, $SD = 2.656$; $p < .001$) was praised for significantly reducing repetitive tasks like script reading (P1, P3, P10), application switching (P12, P13), and typing (P1, P17, P9), and for organizing collected data (P9, P10, P13). This efficiency allowed designers “to focus more on design strategy” (P4) and even “would enable me to interview as many participants as possible because I can now map massive data with GenAI more efficiently than before” (P17).
- **Transforming pain points into opportunities.** Unlike the baseline ($M = 6.30$, $SD = 2.080$) “just skimmed the surface without helping much in digging opportunities” (P3), GeneyMAP ($M = 7.85$, $SD = 1.981$, $p = .048$) directly revealed potential solutions from matching pain points (P15). Participants appreciated this insight, especially as when using the common tool, they often “confused pain points with opportunities” (P11).

		Common Tool		GeneyMAP				
		mean	std	mean	std	t	p	Sig.
Time per JM Output	Time Taken	36.63	4.477	30.72	14.591	1.911	.036	*
	Novelty	5.65	2.298	6.60	2.125	2.472	.016	*
Expert-Assessment on Outputs	Appropriateness	5.20	2.040	5.77	2.134	1.571	.122	–
	Thoughtfulness	5.88	1.776	6.65	1.964	2.395	.020	*
	Interestingness	4.82	2.626	6.02	2.347	2.855	.006	**
	Usefulness	6.27	1.774	6.45	1.872	.598	.552	–
	Surprise	4.93	2.629	6.13	2.548	2.810	.007	**
Self-Assessment on Process	Phase	5.90	2.198	8.10	2.360	-2.674	.015	*
	Mapping	4.55	2.502	8.00	2.656	-4.012	<.001	***
	Opportunities	6.30	2.080	7.85	1.981	-2.111	.048	*
	Inspiration	5.10	2.150	7.75	1.997	-3.251	.004	**
	Completeness	6.00	2.317	7.80	2.142	-2.139	.046	*

Table 5: Results of time taken completing each JM output, CAT expert-assessment on JM outputs, and self-assessment on JM process. (–: $p > .050$, *: $p < .050$, **: $p < .010$, ***: $p < .001$)

- **Aiding in idea iteration.** GeneyMAP ($M = 7.75$, $SD = 1.997$) significantly improved ideation compared to the baseline ($M = 5.10$, $SD = 2.150$, $p = .004$). P1 explained that “GenAI helped me express ideas I could not articulate on my own” while P13 noted “it could find existing works that closely align with my ideas for reference”. Furthermore, P8 stated that it was inspiring to “use GenAI to supplement and validate ideas from an expert perspective, even with just a few keywords.”
- **Improving the completeness of JMs.** Participants preferred GeneyMAP ($M = 7.80$, $SD = 2.142$) significantly over the baseline ($M = 6.00$, $SD = 2.317$, $p = .046$) for its ability to create more comprehensive JMs (P17). P9 stated that “GeneyMAP fully leveraged GenAI’s intelligence to help me avoid overlooking any content”. GeneyMAP was also praised for “polishing and enhancing language, which improved my efficiency” (P10), with P11 adding that “its high completeness stemmed from its well-structured JM template.”

However, participants reported three major concerns about using GeneyMAP, which were attributed to the technological constraints and the extensive involvement of GenAI in the JM process:

- **Inaccurate information mapping.** GeneyMAP struggled to earn users’ full trust due to issues with inaccurate information mapping. Participants expressed concerns such as “the resulting information tended to need a cross-check” (P13) and “GenAI’s inability to grasp the implicit nuances of human instructions. Therefore, I wanted it to directly quote users’ words” (P4). These factors appeared to collectively contribute to the relatively low performance in the *Appropriateness* of JM outputs.
- **Irrelevant design opportunities.** Several participants remarked that some of the generated opportunities were of “insufficient quality and relevance” and characterized some resulting images as “partially chaotic” (P14). P10 noted that the outcomes sometimes “strayed the context.” P16 expressed a significant decline in confidence in GeneyMAP’s ability to generate high-quality design opportunities that perfectly

aligned with the business logic, reflecting a broader decline in *Usefulness* in GeneyMAP’s outputs.

- **Limited design engagement.** As GenAI showed its capability to enhance JM creation efficiency, another issue emerged: “I felt passive and didn’t apply my design thinking” (P7). Some participants even felt that “it took away my pride of being a designer, as I don’t like the way it provided the ideas only for me to revise” (P13). However, P8 argued that “My autonomy as a designer remained intact; GeneyMAP merely helped me handle some of the mundane tasks”. P11 agreed by stating “so far, its output has been rough, and the provided direction was broad, I still needed to engage in independent thinking.”

Several identified strengths of GenAI might explain the statistical findings in JM outputs: *aiding in idea iteration* could be viewed as a key factor leading to the **Novelty** in the JM outputs, and the *improving the JM completeness* intuitively reflected the **Thoughtfulness** of GeneyMAP outcomes. The **Interestingness** and **Surprise** of GeneyMAP outputs could be evidenced in the *comprehensive JM phases distillation* and *transforming pain points into opportunities*. On the other hand, the limitations in *inaccurate information mapping*, *irrelevant design opportunities*, and *limited design engagement* may explain why GeneyMAP appeared to be less effective in enhancing **Usefulness** and **Appropriateness** of JM outputs.

6.2 Perception of Creativity Support

Table 6 shows the paired t-test results of participants’ perceptions of creativity support. GeneyMAP scored significantly higher in terms of **Exploration** ($ps \leq .025$) and **Enjoyment** ($ps < .001$) than the baseline, while no significant difference was found in other dimensions. We present the identified strengths from the reflections of the participants:

- **Providing insightful stimuli.** Participants observed that “the freewheeling ideation occurred while interacting with GenAI; seeing its outputs sparked my imagination” (P7), which contrasted with the static whiteboard-like common tools. P9

		Common Tool		GeneyMAP		t	p	Sig.
		mean	std	mean	std			
Exploration	Diversity	5.65	2.231	7.75	1.410	-2.956	.008	**
	Continuity	6.35	2.477	8.20	1.361	-2.439	.025	*
Expressiveness	Creativity	5.75	2.221	6.10	1.997	-.459	.651	–
	Self-express	6.30	2.557	6.95	2.164	-.779	.446	–
Immersion	Task	6.00	2.616	5.50	2.373	.647	.525	–
	Creation	5.90	2.954	5.65	2.581	.273	.788	–
Enjoyment	Utility	6.15	2.007	8.35	1.954	-3.895	<.001	***
	Experience	5.55	1.820	7.95	2.064	-3.915	<.001	***
Effort	Satisfaction	5.95	2.373	7.45	2.038	-1.751	.096	–
	Worth the effort	6.30	2.250	7.65	2.183	-1.571	.133	–

Table 6: Results of self-perceived experience on CSI. (–: $p > .050$, *: $p < .050$, **: $p < .010$, ***: $p < .001$)

highlighted that “GenAI visualized my concepts, at least I had an idea to start.” (P9). Even the mundane outputs from GenAI might still be perceived as inspiring (P13) while P10 remarked, “GenAI provided ideas I wouldn’t have thought of”, showcasing GeneyMAP’s role in fostering creative thinking.

- **Facilitating progressive discussions.** GeneyMAP could effectively organize online data (P14) and explore possibilities (P3), providing a solid foundation for an in-depth discussion with GenAI for creative exploration. This organization supported the continuity of potential ideas through interactive and in-depth communication with GenAI. P11 highlighted that “GenAI could help me refine and adjust the prompt”, leading to high-quality conversations and better alignment with specific design goals (P9).

While GeneyMAP demonstrated clear strengths regarding the design stimuli, two notable challenges were frequently mentioned:

- **Lacking originality.** The overall visualized opportunities were reliable but not very creative (P10). P14 added that the outputs were “with insufficient originality” and P7 critiqued the generated solutions “could not express my ideas like common tools did, where I would actively filter out valueless thoughts.”
- **Demanding effective prompts.** GeneyMAP’s performance heavily relied on the designers’ ability to craft well-constructed prompts (P12). If the designer was “not sufficiently familiar with prompt construction”, it would “be difficult to clearly describe specific requirements to GenAI” (P1), and “led to repetitive attempts without getting the desired answers” (P8).

6.3 Perception of Agency

According to Table 7, the participants significantly preferred GeneyMAP over the baseline in terms of system **Collaboration** ($p < .001$) and **Adherence** ($p < .001$). However, no significant difference was observed in the dimensions of **Goal Achievement**, **Transparency**, **Trust** and **Control**. We present participants’ rationales and use patterns for the different dimensions in the following sections. We first present the identified strengths of GenAI as below:

	Common Tool		GeneyMAP		t	p	Sig.
	mean	std	mean	std			
Goal	6.45	2.164	7.80	1.989	-1.733	.099	–
Transparency	7.15	1.954	6.45	2.164	.899	.380	–
Control	7.55	2.417	7.75	1.743	-.263	.795	–
Trust	6.90	1.944	6.50	2.351	-.487	.632	–
Collaboration	5.30	3.181	8.55	1.146	-4.951	<.001	***
Adherence	6.35	1.461	9.20	1.105	-7.025	<.001	***

Table 7: Results of self-perceived experience on Agency. (–: $p > .050$, *: $p < .050$, **: $p < .010$, ***: $p < .001$)

- **Enhancing objectivity.** Compared to individuals, GenAI was perceived as less susceptible to the various biases that could influence human judgment (P13, P11). P1 mentioned that human designers might have intentionally skipped some parts of the interviews, which affected the accuracy of the results. P7 added that “When mapping interview data, we often had some preliminary ideas, leading us to summarize the pain points that aligned with those thoughts rather than sticking to the facts.”
- **Empowering designer authority.** With GenAI integrated into the JM creation, the role of designers shifted from executors to decision-makers (e.g., P11, P13). As P9 explained: “I only needed to steer the direction, provide constructive feedback, and select from the best option generated by GenAI” while P6 described a shift towards supervisory roles. This transformation underscored GeneyMAP’s potential to empower practitioners by taking over execution tasks and enabling them to focus on creative aspects of design (P7).

Nevertheless, the participants also identified limitations in GenAI’s *unexplainable reasoning* (**Transparency**) and *challenges in controlling details* (**Control**). These issues contributed to the instability of *GenAI-dominated output quality*, which might further affect their perception of **Goal Achievement**.

- **Unexplainable reasoning.** The unexplainable nature of GenAI’s decision-making process, commonly referred to as the ‘black box’, frustrated some participants (e.g., P9, P12).

A few participants suggested that “I would need GenAI to explain the reasons behind its conclusions visually” (P11). P13 also voiced concerns about transparency, “I was not confident in the GenAI’s phase reasoning because it didn’t explain its process clearly”.

- **GenAI-dominated output quality.** The quality of GenAI-generated outputs was significantly influenced by the capabilities of the GenAI models (P15). P10 criticized the unpredictability, stating “the generated solutions were hit or miss.” Despite these concerns, GeneyMAP’s potential was acknowledged, with P7 expressing a positive outlook: “I would prefer to use it more if the GenAI models became more stable.”
- **Challenges in controlling details.** P11 expressed frustration with the unpredictability, stating, “I couldn’t fully control content generation, and I could not tell if the next generated output would be better than the last.” In contrast, P10 focused on the lack of precision, sharing, “when I requested an image with three legs, GenAI couldn’t deliver accurately. The level of control over GeneyMAP felt weak, making it difficult to steer the output.”
- **Over-reliance on GenAI.** The convenience of GeneyMAP also led to concerns about mental inertia, such as “It posed some risks of not conducting analysis myself because it was too convenient” (P9) and “I found myself becoming dependent on GeneyMAP already” (P12). P13 also warned against over-reliance on GenAI, “I found myself reluctant to think proactively, as it’s hard to transition from luxury to austerity.”

6.4 Perception of Usability

After the paired t-test, the results of NASA-TLX (Table 8) indicated that the GeneyMAP was significantly better than the baseline in terms of **Mental, Physical, Temporal, Performance, and Frustration** ($ps \leq .020$), which might be explained by the above-mentioned strengths (e.g., *Efficient user data mapping* and *Improving the JM completeness*). However, no significant difference was observed between the two conditions regarding the perceived **Effort**. Firstly, we discussed some additional insights to support the results:

	Common Tool		GeneyMAP		t	p	Sig.
	mean	std	mean	std			
Mental	8.00	1.919	4.50	2.236	6.203	<.001	***
Physical	6.15	2.277	2.40	1.635	6.978	<.001	***
Temporal	7.00	2.596	2.65	2.084	5.816	<.001	***
Effort	5.35	2.159	5.60	2.210	-.350	.730	–
Performance	7.35	1.954	4.15	2.519	5.915	<.001	***
Frustration	5.05	2.373	3.25	2.221	2.538	.020	*

Table 8: Results of self-perceived experience on NASA-TLX. (–: $p > .050$, *: $p < .050$, **: $p < .001$)

- **Aligning with expectations.** The workflow of GeneyMAP aligned well with designers’ expectations for GenAI functions, including “features such as elements chosen, data categorization, and ideation generation, which I had anticipated

as likely inclusions” (P6). This alignment made the tool intuitive to use, with its familiarity with common tools (P11) and seamless integration into workflows. P9 remarked that “GeneyMAP can be used directly without extra effort to adapt.” Furthermore, GeneyMAP consolidates the complex tasks into a simplified JM workflow (P1, P5). It was described as “a tool that integrated all the desired GenAI functions for JM creation” (P13).

- **Enabling flexible modification.** Flexibility in modifying outputs was another critical function that enhanced usability (P1). P9 appreciated GeneyMAP for it was “very accommodating, allowing modifications without resistance, which smooths the mapping process.” P12 also valued the fact that “I could decide when and how to use GenAI”. P16 also shared the satisfaction: “Compared with other GenAI tools, the ability to tweak and refine GenAI suggestions easily makes it an invaluable part of my design toolkit”.
- **Improving communication clarity.** Participants found that GeneyMAP facilitated communication as it “could generate intuitive images as design work primarily relied on visual media” (P15). P9 added, “the images perfectly captured my ideas, which was ideal for someone like me who didn’t want to spend effort on visualization”. Besides visualization, GeneyMAP could also “construct a common understanding and eliminate cultural misunderstandings with its use of a universal language” (P14).

Although GeneyMAP was praised for improvements in most dimensions of usability, the perceived **Effort** remained similar to the baseline. Participants also highlighted the GeneyMAP’s limitations of requiring manual verification for unsatisfied output, suggesting potential improvements for future versions of GeneyMAP.

- **Requiring manual verification.** “Though GenAI halved the time for generating the content, I anticipated more time would be spent on verification afterward” (P21). This issue appeared more frequently among designers who initially had little trust in GenAI (P20, P17). P6 elaborated on the increased effort required, stating, “I ended up doing the task twice because I needed to understand both the interview scripts and GenAI’s output”.
- **Refining unsatisfied output.** Some participants would repeatedly “refine the GenAI-generated output using the auto-generate function” (P12), akin to seeking a lucky draw, until they achieved satisfaction. Meanwhile, other participants like P8 “did not get enough information from the GenAI-generated content, and needed to adjust the cart images using the prompt”.

7 DISCUSSION

7.1 Strengths and Weaknesses of Leveraging GenAI in JM Creation

GeneyMAP demonstrates notable efficiency in the JM creation process but inevitably faces challenges regarding the quality of the outputs. We reflect on the strengths and weaknesses drawn from participants’ perceptions and the insights presented in Sections 6.1,

6.2, and 6.4, discussing the trade-off between efficiency and quality when leveraging GenAI in JM creation.

GenAI has shown substantial strengths in accelerating the JM creation process and providing inspiring insights, showcasing its remarkable efficiency. On the one hand, GeneyMAP offers an intuitive workflow that *aligns with designers' expectations*, enhancing *comprehensive JM phases distillation*, *efficient user data mapping*, which *improves the completeness of JMs*. On the other hand, GeneyMAP provides structured inspiration tailored to *transform pain points into opportunities* and can *aid in idea iteration*. By automating complex mapping and visualization tasks, GeneyMAP significantly reduces the time and effort required for JM creation, enabling designers to focus on high-level decision-making.

While GeneyMAP enables a quick overview of user data, *offers insightful stimuli* and can *facilitate progressive discussions*, designers' *limited design engagement* in thoroughly understanding user data may result in JM outputs that lack depth and nuance. Participants also identified issues such as *inaccurate information mapping* and the generation of *irrelevant design opportunities*, illustrating that letting GenAI generate synthetic insights not only risks of *lacking originality* but also may compromise the fundamental principles of user-centered design. Furthermore, some participants noted that while *manual verification* was necessary to *refine unsatisfactory outputs*, generating high-quality content with GenAI also *demand effective prompts*. These challenges were particularly evident in complex design contexts, where nuanced understanding and creative depth are critical.

The trade-off between efficiency and quality was acknowledged by many participants, whose attitudes varied significantly. Designers with positive prior experiences with GenAI (e.g., P3, P17) displayed a higher trust in its analytical capabilities. Convinced of GenAI's comprehensive and reliable analyses, they tended to spend less time verifying the outputs against original user data, quickly completing JM creation with high satisfaction in output quality. In contrast, participants such as P6 and P8, who distrusted GenAI's ability to accurately infer user intentions, preferred manual methods for a more detailed analysis. The manual adjustment for GenAI-generated content was often more time-consuming than that of the common tool, highlighting how individual trust levels and previous experiences with GenAI influence perceptions of its efficiency and quality. To sum up, the utility of GeneyMAP is context-dependent, and the efficiency gains do not universally guarantee high-quality outputs.

7.2 Challenges regarding Agency of GenAI in JM Creation

Regarding the agency of GenAI, GeneyMAP demonstrated its utility in *empowering designer authority* and *enhancing objectivity*, with quantitative results showing that participants recognized their collaboration with the GeneyMAP as distinct from the common tool. Derived from our findings from Section 6.3, significant challenges persist. In particular, **synthetic data risk** relates to GeneyMAP's lower scores in *Trust* and *Goal Achieving*, **transparency issues** correspond to weaker results in *Transparency*, and **role-shifting of designers** aligns with lower ratings in *Control*.

The primary challenge lies in *the risks associated with synthetic data*. In GeneyMAP, while real user JMs are derived from real-world interviews, the “*synthetic user*” function offers an alternative by generating fictional JMs based on predefined scenarios. That is to say, the synthetic JMs stem from GenAI's fabric data rather than authentic user data. Given that user-centered design relies on deep engagement with genuine user insights, synthetic data carries inherent risks of lacking authenticity and nuance. The over-reliance on synthetic data risks detaching designers from authentic user contexts, diminishing empathy, and potentially leading to critical failures in design projects. From studies on existing GenAI-empowered persona [57] and storyboard, we observe that their reliance on synthetic data to generate design inspiration often limits contextual relevance. However, GeneyMAP strives to integrate real-world user data directly into JM structures to ground the inspirations more firmly in authentic user insights, enhancing the consistency of the outputs. It is important to note that GeneyMAP represents our preliminary efforts and the challenges posed by synthetic data warrant further exploration.

Another critical challenge is the *issue of transparency*, particularly regarding how GenAI tools process user interview data and generate opportunities. The lack of transparency notably affects designers' trust in and control over GenAI [6]. Existing literature highlights that transparency is a significant barrier to trust in GenAI systems [9]. In our study, while some participants appreciated GeneyMAP's attempt to clarify the logical steps behind its outputs, they felt it was insufficient. The transparency issue also hinders control over the final outputs, as evidenced by its impact on the rating dimensions of *Appropriateness* and *Usefulness* regarding JM outputs. Without adequate reasoning or effective feedback, designers struggled to adjust the details of JM outputs, making it difficult to reach optimal design solutions.

The third aspect is the *shifting role of designers during the JM creation*, which has been a recurring theme in past research on GenAI in design. Aligning with the identified strength of GeneyMAP (i.e. *enhancing objectivity*, *empowering designer authority*), previous studies have identified that GenAI is perceived as a collaborator rather than a tool when collaborating with humans [53]. However, there were concerns about the potential for over-reliance on GenAI's automation. Some participants felt that automating key aspects of JM creation reduced their hands-on involvement in the design process. The over-reliance issue was especially noticeable when GenAI-generated outputs required minimal editing, leading designers to feel disconnected from the creative process.

To address the above concern, future design education could deepen designers' understanding of how to integrate GenAI into the JM creation process thoughtfully [12], ensuring that designers are able to critically evaluate and refine GenAI-generated outputs. By training designers to maintain an active and decisive role, they can enhance their ability to navigate GenAI-empowered tools without becoming overly reliant on them. At the same time, the development of designer-friendly GenAI tools becomes crucial in this educational context. The future tools are supposed to enable designers to effectively understand and apply GenAI in their work without requiring a deep dive into the underlying algorithms. The implications of how to develop such designer-friendly tools will be thoroughly discussed in Section 7.3.

7.3 Implications for Designing Future GenAI-empowered JM and Design Tools

Drawing from the development and evaluation of GeneyMAP, we reflect on the implications for designing future GenAI-empowered design tools, which expands upon our design goals identified in the formative study. To maintain GenAI's efficiency while ensuring quality outputs, we advocate for further advancement of the "human-in-the-loop" paradigm, which integrates designers' active involvement to validate, refine, and contextualize GenAI outputs. The engagement allows designers to leverage automation for efficiency while maintaining creative control to ensure quality. Grounded in this advocacy, we derive the following design implications to guide the development of future GenAI-empowered JM and design tools.

First, retain the common JM creation or design workflows while incorporating GenAI applications. Designers prefer GenAI tools that align with familiar processes so the learning curve can be minimized, as demonstrated by the appreciation with *D1-Distill Templates* in GeneyMAP, strengths in "aligning with expectations" and the received high ratings among the JM process (i.e. *Phase, Mapping, Opportunities, Inspiration* and *Completeness*). The approach of retaining design workflow has also proven effective in GenAI-empowered persona [43], which integrates automation into common creation workflow of user profiles, maintaining familiarity while boosting efficiency. Future GenAI design tools would benefit from streamlining the design workflow instead of disrupting it, thereby enhancing designer engagement.

Second, support prompt writing tailored to JM or design tasks. The lack of prompt writing skills would limit the originality of the GenAI-generated content [70], and general online forums like PromptBase (<https://promptbase.com>) may not be sufficient for assisting specific JM tasks. Future GenAI-empowered JM tools are expected to provide tailored templates, interactive refinements, and example-based guidance to create effective prompts for tasks like defining journey phases or merging user maps. For instance, GenAI-empowered storyboard [39] has demonstrated how integrating context-specific templates can streamline scenario visualization, enabling designers to create relevant and detailed outputs. Similarly, integrating such customized templates into other design tools can help designers get the most out of GenAI for a given design task [40], further advancing *D6-Language Enhancement* to lower the barrier to be engaged in GenAI workflow.

Third, enable GenAI to take on diverse roles throughout the JM creation or design process. Current GenAI plug-ins [51] in platforms like Figma primarily focus on isolated design tasks, without facilitating the simulation of multiple stakeholders. Future GenAI-empowered JM tools could take on roles such as design experts for conducting user interviews, researchers for data analysis, and co-designers for evaluating solutions. Additionally, simulating stakeholders like typical users can offer valuable, fresh perspectives. The dynamic GenAI- designer collaboration, tied to *D5-Stakeholders Simulation*, enhances JM tools by enabling seamless role adaptation to suit specific design tasks while fostering proactive engagement with diverse stakeholder perspectives.

Fourth, validate the synthetic data before use. The design community is supposed to develop actionable guidelines for when and how synthetic data should be used in the design process. For

example, designers should reserve synthetic data for initial prototyping or situations where user data is unavailable or unethical to collect. When synthetic data is necessary, designers could develop and adhere to strict validation procedures to ensure the accuracy and reliability of synthetic data. GenAI-empowered usability evaluation tools like SimUser [71] validate synthetic usability feedback by cross-referencing it with real-world evaluation data to ensure reliability. If not carefully curated and validated, synthetic data can lead to misleading insights or reinforce existing biases, ultimately compromising the validity of design outcomes.

To address the challenges related to the transparent issue, we **propose establishing mechanisms for transparent GenAI-aided design process.** HCI researchers have introduced methods such as identifying information sources [72] and providing external knowledge support [2]. Based on our findings, we suggest that the step-by-step breakdown of logical chains explaining GenAI decisions can help designers understand the rationale behind GenAI's decisions. Moreover, designers seek accountability or traceability of GenAI outputs to strengthen the trust in GenAI. By linking GenAI-generated insights to their sources (e.g., user quotes or interview data), designers can verify authenticity and understand the context. Lastly, designers demand highly controllable generated content leading to insightful design opportunities. Future tools ought to prioritize controllability and flexibility of the generation process, allowing designers to better align the generated content with their creative vision (supports *D3-Opportunities Identification*).

7.4 Limitations and Future Work

A significant limitation of this study is the demographic profile of the participants, predominantly from mainland China or Hong Kong. Considering that work practices are often influenced by socio-cultural contexts [13], our conclusions drawn about the integration of GenAI in JM creation processes might not be directly transferable to different cultural settings. Besides, in designing GeneyMAP, a significant limitation is its exclusive support for text-based input (i.e., interview scripts), neglecting other types of data sources, although GeneyMAP is integrated with GenAI's multi-modal function. Moreover, specific GenAI functions like extractive summarization (i.e., directly extracts user quotes instead of generating abstractive summaries) and efforts to merge individual JMs using various logic, remain largely unexplored. These limitations may limit the tool's generalizability across the entire design process, and hinder the exploration of how GenAI could empower designers.

However, our study provides a valuable case for applying GenAI in JM practices and contributes empirical knowledge and design insights for integrating GenAI in design tools. Future research could focus on leveraging GenAI's multi-modal capabilities to analyze the pictures, audio, and video, as well as present analysis results in a multi-modal format. Also, it is reasonable for future research to increase the adaptability of GeneyMAP by addressing the limitations (e.g., *demand effective prompts* and *challenging in controlling the details*) identified in this study.

8 CONCLUSION

This paper introduced GeneyMAP, a tool designed to enhance the JM creation process by integrating GenAI functions. Six design

goals were found to meet designers' expectations of integrating GenAI into the JM creation process during a 24-participant formative study. The design of GeneyMAP was guided by the design goals. Then, our user study, involving 20 participants, provided valuable insights into the strengths and challenges of using GeneyMAP compared to a common JM tool. Participants reported that GeneyMAP significantly improved the efficiency of user journey mapping, enabling more dynamic exploration of design opportunities (RQ2). In terms of the perception of the GeneyMAP, the participants appreciate its efficiency and utility. However, designers also expressed concerns about the potential loss of control over the design process and the need for constant verification of GenAI-generated data. Despite these challenges, our findings highlighted GeneyMAP's potential to refine JM practices by providing powerful data-analytic and visual support that enhances both the productivity and creativity of designers (RQ3). Reflecting on these insights, our study emphasizes the challenges in synthetic data use, the shifting role of designers, and the transparency issue. We further suggest the implications for further development of GenAI tools in JM creation and UX design, with a focus on balancing GenAI automation with a human-in-the-loop approach, improving transparency by logically displaying the reasoning behind GenAI decisions.

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APPENDIX - A

No.	Gender	Age	Design Exp.	Education Background
I1	M	28	9 years	Media Art Design
I2	M	28	8 years	Industrial Design
I3	M	27	8 years	Industrial Design, UX Design
I4	M	32	8 years	Product Design
I5	M	26	7 years	Industrial Design
I6	M	27	7 years	Interaction Design, UX Design
I7	F	28	11 years	Product Design
I8	F	27	9 years	Interaction Design, Graphic Design
E1	M	34	16 years	Product Design
E2	M	30	12 years	Intelligent Product Design
E3	M	33	16 years	Intelligent Product Design
E4	M	35	15 years	Digital Art Design
E5	F	34	15 years	Industrial Design, Interaction Design
E6	M	34	13 years	Industrial Design
E7	F	28	10 years	UX Design
E8	M	32	13 years	Interaction Design, UX Design
S1	F	23	5 years	Intelligent Product Design, UX Design
S2	M	22	5 years	Industrial Design
S3	F	23	6 years	Industrial Design
S4	F	25	5 years	Industrial Design
S5	M	24	4 years	Industrial Design
S6	M	23	5 years	Industrial Design, UX Design
S7	F	21	4 years	Product Design
S8	F	23	4 years	Industrial Design

The demographic information of the participants recruited for formative study.

APPENDIX - B

Common Tool Interface

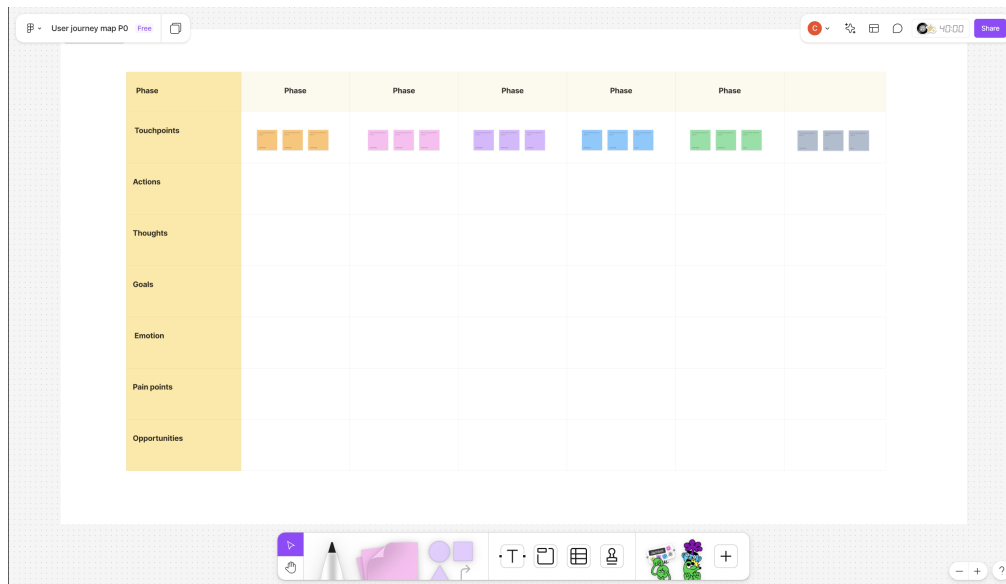


Figure 7: The JM template provided by Figma.

APPENDIX - C

Interview Scripts for User Study

Scripts Example - 1 (Script I)

.....

Interviewer: “How Do you usually plan your trip home after work?”

Maria: “I usually pick up my car in the company parking lot after work.”

Interviewer: “What do you usually do while driving?”

Maria: “Usually, I turn on the music or the radio, and sometimes I talk to people in the car. If no one comes home with me, I listen to some news radio or play some light music to keep me awake. I usually choose light music or pop songs, which have a strong sense of rhythm, more refreshing. If it is radio, I like to listen to the news and talk shows, rich content, can attract my attention, not easily bored.”

.....

Scripts Example - 2 (Script I)

.....

Interviewer: “What do you think about on the way to the garage? Can you describe your mood?”

Samir: “I usually relax and change my mood on that part of the trip. I think back to important moments of the day and maybe plan my evening activities. I usually feel both expectant and a little tired.”

Interviewer: “How Do you usually feel when you get out of the car?”

Samir: “When I get out of the car, I usually feel a sense of relief. It’s always nice to come home, especially after a busy day, to be able to relax and be with my family, which I cherish.”

.....

Scripts Example - 3 (Script II)

.....

Interviewer: “What do you do after browsing and selecting a product? Any unexpected problems or situations? Do you think it is convenient to take out the items from the shopping cart and scan the code?”

John: “I’ll check out at the cashier later. It’s relatively easy to scan items out of a shopping cart, and there are no surprises. We usually work with the salesperson to transfer the items from the shopping cart to the table.”

Interviewer: “What problems and difficulties will you encounter after you finish the checkout?”

John: “Sometimes you can’t find a place to return your cart, or people’s carts are scattered all over the place. But this should be a question of supermarket management”

APPENDIX - D

D.1 Common Tool Outputs Example

Phase	Preparation at Home	旅途中	抵达超市	购物中行进	购物拿取	结账
Touchpoints	前置购物清单 小推车	前置购物清单	前置购物清单, 购物车	购物车	商品陈列架	收银台, 收银
Actions	1. 前置购物清单, 扫描前置购物清单 2. 前置购物清单, 扫描前置购物清单	前置购物清单	前置购物清单, 扫描前置购物清单, 工作人员扫码支付 前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单	前置购物清单
Thoughts	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单
Goals	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单 前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单
Emotion	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单
Pain points	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单
Opportunities	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单	前置购物清单, 扫描前置购物清单

Figure 8: Common tool output of Script II - P10

D.2 GeneyMAP Outputs Example

	1 发现	2 选择	3 准备	4 整理物品	5 使用工具
TouchPoint	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单
Action	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单
Thought	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单
Emotion	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单
PainPoint	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单	前置购物清单 前置购物清单 前置购物清单

Figure 9: GeneyMAP output of Script II: part 1 - P2



Figure 10: GeneyMAP output of Script II: part 2 - P2

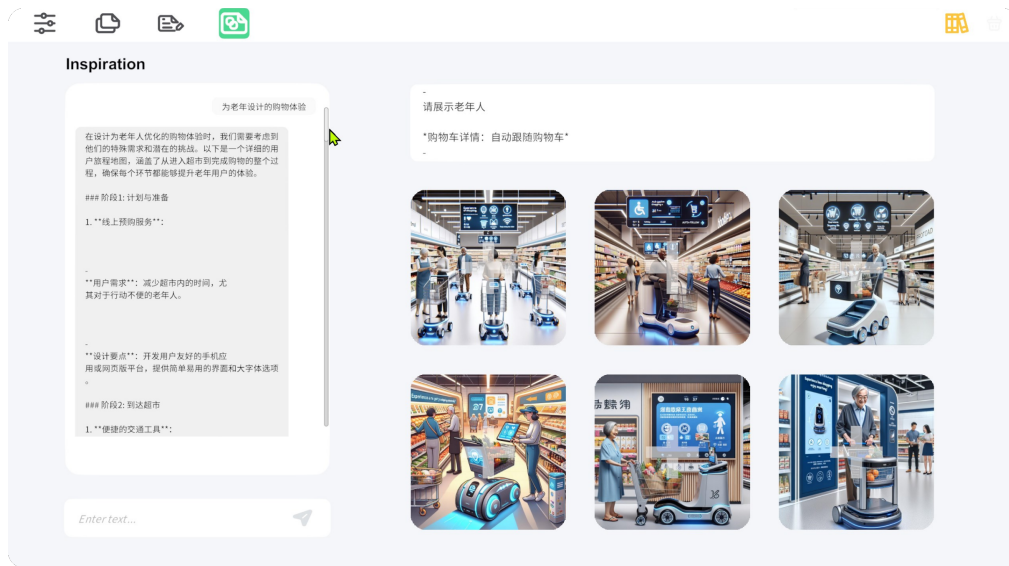


Figure 11: GeneyMAP output of Script II: part 3 - P2